Dynamic Panels

BUSS975: Causal Inference in Financial Research

Professor Ji-Woong Chung

Korea University

Where we left off

- Fixed effects rests on one idea: anything about a firm that **does not change** cannot explain why its outcome changed. The price was **strict exogeneity**: today's shock must not feed into tomorrow's regressors.
- Leverage, payout, cash, investment all depend on **their own past**. Put last year's Y on the right-hand side and strict exogeneity is not doubtful — it is **false by construction**.

- One question runs through this lecture:

  Is a firm high this year because it was high last year —
  or because it is simply a high firm?

- Every estimator we meet is a different way of answering it. The two you already know both get it wrong, in opposite directions. That turns out to be useful.

The plan

1. **The problem.** OLS and fixed effects both get persistence wrong — and we know which way.
2. **The fix.** A longer lag of the same variable, used as an instrument.
3. **When the fix fails.** Persistent series, large firm effects — exactly the finance case — and the repair.
4. **In practice.** Leverage adjustment, the code, what to report.

1. The Problem

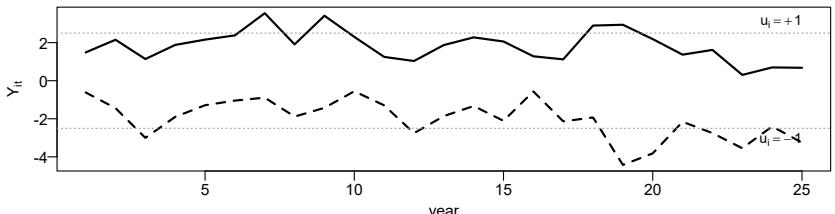
Most things in finance drift

They do not jump to a target. They close part of the gap each year.

$$Y_{it} \;=\; \rho\, Y_{i,t-1} + \delta D_{it} + u_i + \varepsilon_{it}, \qquad |\rho| < 1$$

- ρ **is the economics.** A firm closes a fraction $1-\rho$ of its gap each year; a shock has half-life $\ln 0.5 / \ln \rho$ — 1.4 years at $\rho = 0.6$, 6.6 at $\rho = 0.9$. Whether firms *rebalance* toward a target or merely *drift* is a question about ρ.
- δ is now the **impact** effect; the long-run effect is $\delta/(1-\rho)$. Get ρ wrong and you get that wrong too.

Two firms, one question



Same $\rho = 0.6$, same shocks in distribution, different u_i. Each firm wiggles around its own long-run level $u_i/(1-\rho)$.

- The firm effect sets the **height**. ρ sets **how fast a wiggle dies out**. Those are the two things an estimator must keep apart.
- Pooled OLS reads height as persistence. Fixed effects, removing height, manufactures mean reversion. Next two slides.

OLS reads height as persistence

Strip the model to the bone: set $\rho = 0$. No persistence at all, only firms that differ permanently.

$$Y_{it} \;=\; u_i + \varepsilon_{it}$$

- A high-u_i firm is high this year **and** was high last year. Not because the past caused the present — the firm is simply like that. OLS sees Y following Y:

$$\operatorname{plim} \hat{\rho}_{\mathrm{OLS}} \;=\; \frac{\sigma_u^2}{\sigma_u^2 + \sigma_\varepsilon^2} \;>\; 0 = \rho$$

- Simulated with $\sigma_u = \sigma_\varepsilon$: $\hat{\rho}_{\mathrm{OLS}} = 0.50$ when the truth is **zero**. **Biased up**, and more years do not help — more years of the firm being itself is more of the same evidence.

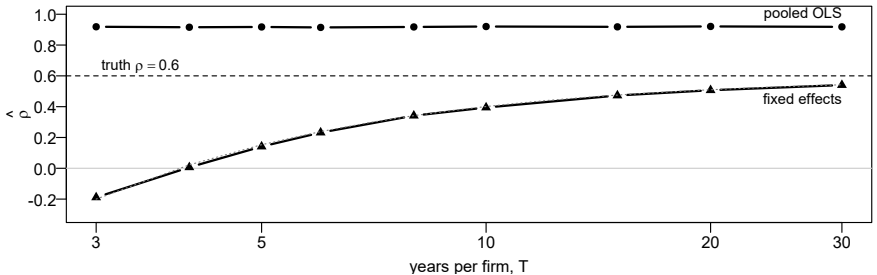
Fixed effects manufactures mean reversion

Demeaning subtracts each firm's *average* error from every year's error. A lucky shock last year raised $Y_{i,t-1}$ — and sits inside that average.

$$\underbrace{Y_{i,t-1} - \bar{Y}_i}_{\text{regressor: contains } +\varepsilon_{i,t-1}} \qquad \underbrace{\varepsilon_{it} - \bar{\varepsilon}_i}_{\text{error: contains } -\varepsilon_{i,t-1}/T}$$

- The year the regressor is high is a year the error has been pushed down. The fitted slope tilts **down**: **Nickell (1981) bias**, of order $1/T$.
- Same $\rho = 0$ experiment, $T = 5$: $\hat{\rho}_{\text{FE}} = -0.25$, which is $-1/(T-1)$. OLS invented persistence; demeaning invents reversion.
- It happens **even without any firm effect**: demeaning alone does it.

Both biases in one picture



$\rho = 0.6$, 4,000 firms, $\sigma_u = \sigma_\varepsilon$. Dotted: Nickell's formula for the fixed-effects plim.

- **OLS never moves.** Heterogeneity is not resolved by more years.
- **Fixed effects heals, slowly.** At $T = 3$ it is *negative* for a process with persistence 0.6; at $T = 30$ it is still short. Finance panels are 10–20 years. The more persistent the series, the slower the bias leaves.

Two wrongs that fence in the truth

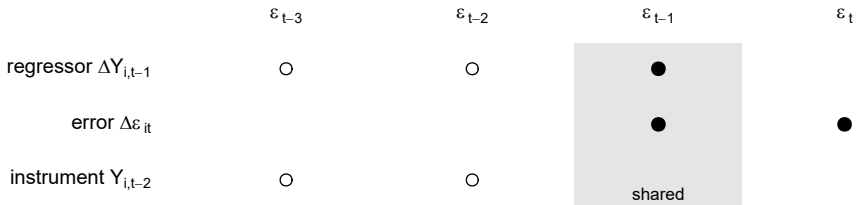
$$\hat{\rho}_{\text{FE}} \;<\; \rho \;<\; \hat{\rho}_{\text{OLS}}$$

- Neither is consistent, but you know the **sign** of each bias. At $T = 5$ above: $0.14 < 0.60 < 0.92$. Wide — and honest.
- **The standard first move** (Bond 2002): run both *before* anything clever, and keep both numbers. They are the goalposts.
- **Any estimator you trust must land between them.** One that lands outside is telling you an assumption failed — before you interpret a single coefficient.

2. The Fix

Differencing kills the firm effect — and shows the real problem

$$\Delta Y_{it} \;=\; \rho\,\Delta Y_{i,t-1} + \delta\,\Delta D_{it} + \Delta\varepsilon_{it}$$



- u_i is gone. But the regressor $\Delta Y_{i,t-1}$ contains last year's shock, and so does the error $\Delta\varepsilon_{it}$ — **with a minus sign**. Differencing is not the fix; it is where the fixed-effects bias comes from.
- **Go back one more year.** $Y_{i,t-2}$ is built from shocks dated $t-2$ and earlier: it shares nothing with the error, yet it is part of $\Delta Y_{i,t-1}$, so it predicts the regressor.
- That is **Anderson–Hsiao (1982)**: IV on the differenced equation, $Y_{i,t-2}$ as instrument. **A broken regressor and a valid instrument are one date apart.**

Arellano–Bond: use every lag you have

At date t, *every* Y dated $t-2$ or earlier is a valid instrument. Use them all (Holtz-Eakin, Newey and Rosen 1988; Arellano and Bond 1991):

Differenced equation at	valid instruments for $\Delta Y_{i,t-1}$	count
$t = 3$	Y_{i1}	1
$t = 4$	$Y_{i1},\ Y_{i2}$	2
$t = T$	$Y_{i1},\ \ldots,\ Y_{i,T-2}$	$T-2$

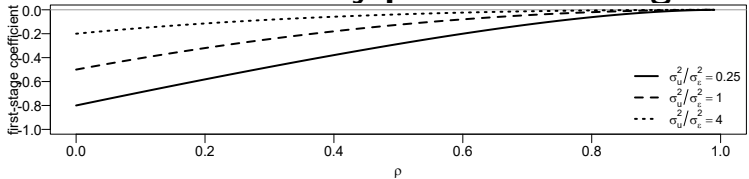
- $(T-1)(T-2)/2$ instruments in all, so the estimator is **GMM**. Gain: Anderson–Hsiao with one instrument is very noisy; this is far tighter.
- Cost: the count grows with T^2 — 91 at $T = 15$, 406 at $T = 30$. **Too many instruments** (Roodman 2009) overfit the regressor, contamination included, and GMM slides back toward OLS on the differenced equation, which is biased *down*. Hansen's J loses the power to object.
- Remedies: **cap** the lag depth, **collapse** the instrument set to one column per lag. Keep the count well below N, and report it.

What to report, and why

- **Windmeijer (2005) standard errors.** Two-step GMM is efficient but its standard errors are far too small in finite samples. One-step robust and two-step Windmeijer should agree; if they do not, something is weak.
- **AR(2) test on $\Delta\varepsilon$.** AR(1) is expected — adjacent differences share $\varepsilon_{i,t-1}$. AR(2) must be *absent*: it appears only if ε itself is serially correlated, and then $Y_{i,t-2}$ shares a shock with the error and **every instrument fails**. This is the test of the identifying assumption.
- **Hansen J**, with the instrument count next to it. J is informative only when the count is small relative to N.
- **The bracket.** Below $\hat{\rho}_{\text{FE}}$: weak or too many instruments. Above $\hat{\rho}_{\text{OLS}}$: a violated assumption — next section shows one.

3. When the Fix Fails

Weak instruments: a level barely predicts a change



Each year a firm closes a fraction $1-\rho$ of its gap to its own long-run level:

$$\Delta Y_{i,t-1} \;=\; -(1-\rho)\,\underbrace{\left(Y_{i,t-2} - \tfrac{u_i}{1-\rho}\right)}_{\text{gap}} + \varepsilon_{i,t-1} \qquad \Longrightarrow \qquad \gamma \;=\; (\rho-1)\,\frac{\text{Var(gap)}}{\text{Var}(Y_{i,t-2})}$$

- The level predicts the change **only through the gap**; its height predicts nothing. **Two ways to die:** $\rho \to 1$, the gap closes too slowly to matter; large σ_u^2, the level is mostly height (Blundell and Bond 1998).
- $\gamma = -0.08$ at $\rho = 0.6$; -0.001 at $\rho = 0.95$. Leverage, payout and cash live in the right-hand third. **The instrument is weakest exactly where you need it.**

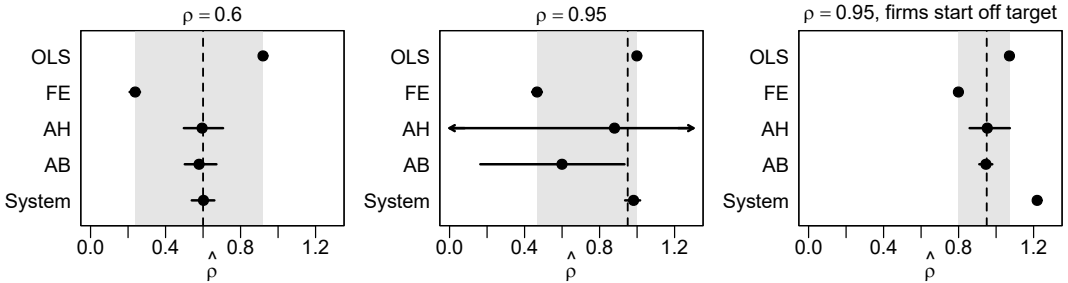
System GMM: instrument the level with the change

Blundell and Bond (1998): keep the differenced equation, and **add the levels equation**, instrumented the other way round:

$$Y_{it} = \rho\, Y_{i,t-1} + u_i + \varepsilon_{it}, \qquad \text{instrument } Y_{i,t-1} \text{ with } \Delta Y_{i,t-1}$$

- **Why it helps.** A level cannot predict a random walk's next step — but **a firm that just rose is high**. The first-stage coefficient of $Y_{i,t-1}$ on $\Delta Y_{i,t-1}$ is $\frac{1}{2}$ for *every* ρ. It never dies.
- **What it costs.** The levels error contains u_i, so a firm's changes must say nothing about its permanent level: $\mathbb{E}[\Delta Y_{i,t-1}\, u_i] = 0$. True in the two-firm picture *because both firms started at their own long-run levels* — **mean stationarity**.
- **When to doubt it.** Firms observed from IPO start away from target and converge; a panel that starts after a crisis has everyone off target together. Then rising *does* reveal a high-u_i firm, and the extra instruments are invalid.

Every estimator, on one line



500 firms, $T = 6$, 200 draws. Dots: medians; bars: 10th–90th percentiles; shaded: the bracket; dashed: the truth.

- $\rho = 0.6$**:** every IV lands in the bracket, near the truth. $\rho = 0.95$**:** the difference estimators collapse; system GMM holds (0.98).
- **Firms start off target:** Arellano–Bond recovers (0.95); system GMM lands at 1.22, *above OLS* — and the bracket catches it. **Each estimator fails exactly when its own assumption does.**

Long panels: a different route

- **Bias-corrected fixed effects.** Nickell's bias has a known formula, so subtract it: estimate FE, correct at $\hat{\rho}$, iterate (Kiviet 1995; Bruno 2005; Stata `xtlsdvc`). No instruments, so nothing to be weak — but the other regressors must be strictly exogenous, which finance controls rarely are.
- **Horse race on finance-like panels** — short, persistent, unbalanced (Flannery and Hankins 2013): bias-corrected FE and system GMM generally reliable; difference GMM frequently not.

If	then
$T \leq 10$, ρ moderate	Arellano–Bond, with the diagnostics
ρ near one	system GMM, with mean stationarity defended
$T \geq 20$, exogenous controls	bias-corrected fixed effects

4. In Practice

Case study: how fast do firms adjust leverage?

The partial-adjustment model is our equation with $Y =$ leverage and $1 - \rho$ the **speed of adjustment**.

Paper	Estimator	$\hat{\rho}$	speed $1 - \hat{\rho}$
Fama and French (2002)	pooled (Fama–MacBeth)	≈ 0.85	$\approx$ 7–18%
Flannery and Rangan (2006)	FE, lagged leverage instrumented	≈ 0.66	$\approx$ 34%
Lemmon, Roberts and Zender (2008)	system GMM	≈ 0.75	$\approx$ 25%
Huang and Ritter (2009)	long differencing	≈ 0.80	$\approx$ 17–23%

- **The ordering is the bracket.** Fixed effects reports the *fastest* adjustment, pooled estimation the *slowest*; the IV-type estimates sit between.
- Whether firms rebalance toward targets or just drift **turned on the bias this lecture is about**: a half-life of 1.7 years and one of 5 are different theories of the firm.
- Caveat: samples, periods and leverage definitions differ across the papers. The ordering is suggestive, not a controlled horse race.

The code

```
* 1. The bracket first -- and keep both numbers
reg   y L.y x i.year, cluster(firm)
xtreg y L.y x i.year, fe cluster(firm)

* 2. Difference GMM (Arellano-Bond): two-step, Windmeijer SEs, lags 2-4 of y
xtabond y x i.year, lags(1) maxldep(3) twostep vce(robust)
estat abond                          // AR(1) expected; AR(2) must be absent

* 3. Same, with collapsed instruments and the Hansen test (Roodman's xtabond2)
xtabond2 y L.y x i.year, gmm(y, lag(2 4) collapse) iv(x i.year) ///
         noleveleq twostep robust
*    x predetermined rather than exogenous? iv(x ...) -> gmm(x, lag(1 3) collapse)

* 4. System GMM -- only after you have defended mean stationarity
xtdpdsys y x i.year, lags(1) maxldep(3) twostep vce(robust)

* 5. Long T, exogenous x: bias-corrected fixed effects
xtlsdvc y x, initial(ab) bias(3) vcov(50)
```

- R: `plm::pgmm()` with `transformation = "d"` or `"ld"`. Simulation to grade them against a known ρ: `06b_dynamic_panel_sim.do` (one draw in the backup).

Conclusion

1. **A lagged Y is endogenous by construction.** The question is whether a firm is high because it *was* high, or because it *is* high.
2. **OLS says "was high" too often; fixed effects too rarely.** Run both first: **the bracket is the goalposts.**
3. **The fix is a longer lag of the same variable.** $Y_{i,t-2}$ is one date too early to be contaminated. Arellano–Bond uses all such lags; cap and collapse them.
4. **The fix fails when the series is persistent or firm effects are large** — the normal case in finance. Then system GMM, if firms sit at their own long-run levels; or bias-corrected FE, if T allows.
5. **Each estimator fails exactly when its own assumption does**, and the bracket usually shows it. Ask of every estimate: *which way is it wrong?*

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Backup

B1. The date rule, and the Nickell bias at $T = 3$

Y_{is} is built from u_i and shocks dated s and earlier. If ε is serially uncorrelated and unrelated to u_i:

$$\mathbb{E}[Y_{is}\,\varepsilon_{it}] = 0 \;\text{ for } t > s, \qquad \mathbb{E}[Y_{is}\,\varepsilon_{is}] = \sigma_\varepsilon^2$$

With $T = 3$ there is one differenced observation per firm, so $\hat{\rho}_{\mathrm{FE}}$ is least squares on $\Delta Y_{i3} = \rho\,\Delta Y_{i2} + \Delta\varepsilon_{i3}$:

$$\operatorname{plim}(\hat{\rho}_{\mathrm{FE}} - \rho) = \frac{\mathbb{E}[\Delta Y_{i2}\,\Delta\varepsilon_{i3}]}{\mathbb{E}[(\Delta Y_{i2})^2]} = \frac{-\sigma_\varepsilon^2}{2\sigma_\varepsilon^2/(1+\rho)} = -\,\frac{1+\rho}{2}$$

- Numerator: expand $(Y_{i2} - Y_{i1})(\varepsilon_{i3} - \varepsilon_{i2})$. Three products pair a Y with a *later* shock: zero. One pairs Y_{i2} with its own shock, with a minus sign: $-\sigma_\varepsilon^2$.
- Denominator: u_i cancels in a difference; what is left has variance $2\sigma_\varepsilon^2/(1+\rho)$.
- The plim is negative for every $\rho < 1$. (Hansen 2022, §17.37.)

B2. One date apart, as a table

Each cell is $\mathbb{E}[\text{row} \times \text{column}]$; the last column is the first minus the second.

	ε_{it}	$\varepsilon_{i,t-1}$	$\Delta\varepsilon_{it}$
$Y_{i,t-1}$	0 (later shock)	$\boldsymbol{\sigma_\varepsilon^2}$ (same date)	$-\sigma_\varepsilon^2$
$Y_{i,t-2}$	0 (later shock)	0 (later shock)	0
$\Delta Y_{i,t-1} = Y_{i,t-1} - Y_{i,t-2}$			$\boldsymbol{-\sigma_\varepsilon^2}$

- **Validity.** The regressor inherits the one same-date term; the instrument, one year older, has none.
- **Relevance.** $\text{Cov}(Y_{i,t-2}, \Delta Y_{i,t-1}) = -\sigma_\varepsilon^2/(1+\rho) \neq 0$ — but small when ρ is near one, which is the weak-instrument problem of Section 3.
- **The zero row needs serially uncorrelated ε.** If the truth is AR(2) and you fit AR(1), $\varepsilon_{i,t-2}$ inside $Y_{i,t-2}$ predicts $\varepsilon_{i,t-1}$ and the instrument is invalid. That is what the AR(2) test checks.

B3. Try it: simulate, estimate, grade

```
clear all
set seed 20260202
local rho = 0.6                          // then 0.95
set obs 500
gen firm = _n
gen u = rnormal()                        // firm effect
* start each firm at its own long-run level
gen y = u/(1-`rho') + rnormal(0, 1/sqrt(1-`rho'^2))
expand 8
bysort firm: gen year = _n
xtset firm year
by firm: replace y = `rho'*y[_n-1] + u + rnormal() if _n > 1

reg   y L.y, vce(cluster firm)                    // up
xtreg y L.y, fe vce(cluster firm)                 // down
ivregress 2sls D.y (LD.y = L2.y), vce(cluster firm)
xtabond  y, lags(1) maxldep(3) twostep vce(robust)
xtdpdsys y, lags(1) maxldep(3) twostep vce(robust)
```

One draw, $\hat{\rho}$:

	0.6	0.95	0.95^{off}
OLS	0.918	0.999	1.083
FE	0.334	0.592	0.840
AH	0.596	1.077	1.004
AB	0.609	0.824	0.966
System	0.589	1.010	**1.150**

Full file: `06b_dynamic_panel_sim.do`. Set `on_target = 0` to start firms at the average level (^{off}).

One draw is noisy — read it against the figure in Section 3.

B4. A reporting checklist

Report	Because
$\hat{\rho}_{\text{OLS}}$ and $\hat{\rho}_{\text{FE}}$	the bracket; your estimate must sit inside it
AR(1) and AR(2) tests	AR(2) tests the identifying assumption
Hansen J, the instrument count, lag depth, collapsed or not	J is informative only when the count is small relative to N
one-step and two-step (Windmeijer) side by side	disagreement signals weakness
difference vs. system GMM, with mean stationarity defended	they rest on different assumptions
what D is assumed to be (exogenous / predetermined / endogenous)	it determines which lags of D are valid instruments

B5. Bracketing δ: lagged Y or fixed effects?

When you are unsure **which model** is right — lagged Y, or fixed effects — and what you care about is δ, the two estimates bracket it too.

If the truth is	and you estimate	then $\hat{\delta}$ is
the lagged model	fixed effects	too **high**
the fixed effects model	the lagged model	too **low**

- Intuition: under the lagged model, a treated firm's $Y_{i,t-1}$ already carries part of the effect, and fixed effects, ignoring it, attributes that part to D. Under the fixed-effects model, the lagged Y absorbs part of the firm effect *and* part of the treatment.
- Angrist and Pischke (2009, §5.4): report both. If they agree on the sign and roughly on the size, the question of which model is right matters less than you feared.